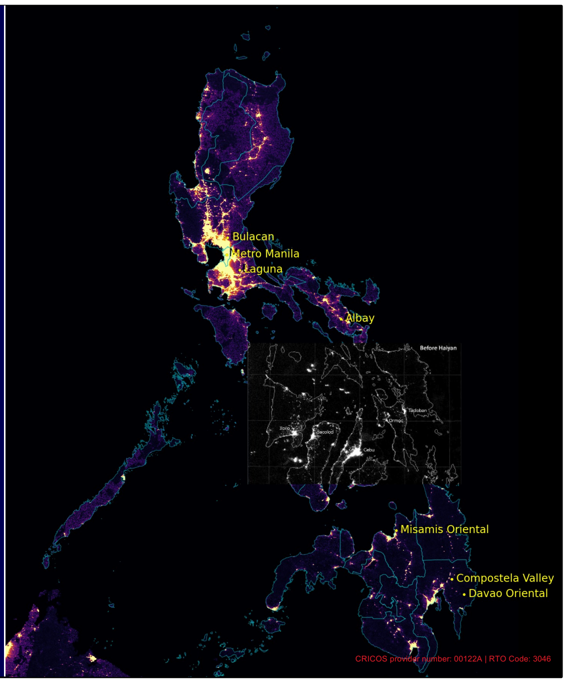
Monthly Meeting

RMIT-UT-ITC Cotutelle (Joint PhD) Collaboration

Project Title: Reliability-Qualified Nighttime Lights for Disaster Impact, Recovery, and Resilience Analysis

Candidate:
Rene L. Principe Jr.

Supervisory Panel:
Prof. Simon Jones (RMIT University)
Assoc. Prof. Karin Reinke (RMIT University)
Prof. Norman Kerle (University of Twente – ITC)
Assoc. Prof. Luigi Lombardo (University of Twente – ITC)



This presentation brings together three pieces of work completed over the past month: the review of related literature, the emerging RQ2 concept, and the first exploratory analysis of the Haiyan case.

The proposed direction is still open. My aim today is to explain how the literature has shaped my thinking, identify several gaps that may offer a meaningful direction for RQ2, and show preliminary observations that help determine which questions are worth pursuing.

I will end with the areas where I would particularly value the panel’s advice, especially on the treatment of viewing geometry, the relationship with the existing Haiyan recovery research, and the appropriate scope of the NTL interpretation.

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RMIT UNIVERSITY

Date: August 6, 2026

Meeting Agenda:

1. Literature synthesis,
2. emerging research opportunities,
3. preliminary analysis of Haiyan/Yolanda

Questions for discussion:

- **Direction:** Is metric identifiability from reliability-qualified NTL a strong RQ2 contribution?
- **Scope:** Which parts should be core, and which should remain sensitivity or appendix?
- **Integration:** How should NTL functional recovery be positioned against existing Haiyan physical-recovery work?

Today I will briefly summarize the literature, explain the research opportunity I see, and show the first round Haiyan analysis.

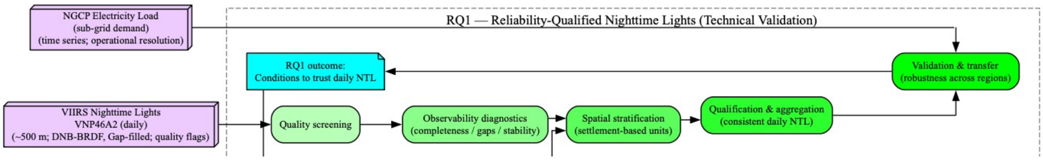
I want to be clear that RQ2 is still being shaped. I am mainly seeking advice on three things: whether metric identifiability is a strong direction, how tightly the scope should be defined, and how the NTL work should connect with existing Haiyan physical-recovery studies.

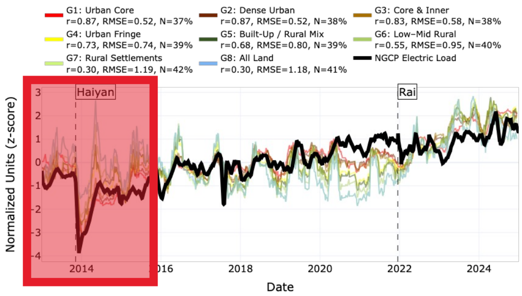
The target audience/journal for this is also something I am trying to reconcile, because there are times when it veers to become a technical one, and it's always trumped with why does it matter, which swerves the direction to prioritize application.

By the end of thus meeting, I'd like to get insight wether the proposed direction is reasonable and worth developing further, and which direction.

Starting point: RQ2 grows from the observation problem examined in RQ1

RQ1: When can a daily VIIRS Black Marble observation be treated as sufficiently reliable for analysis?





Why is this important?

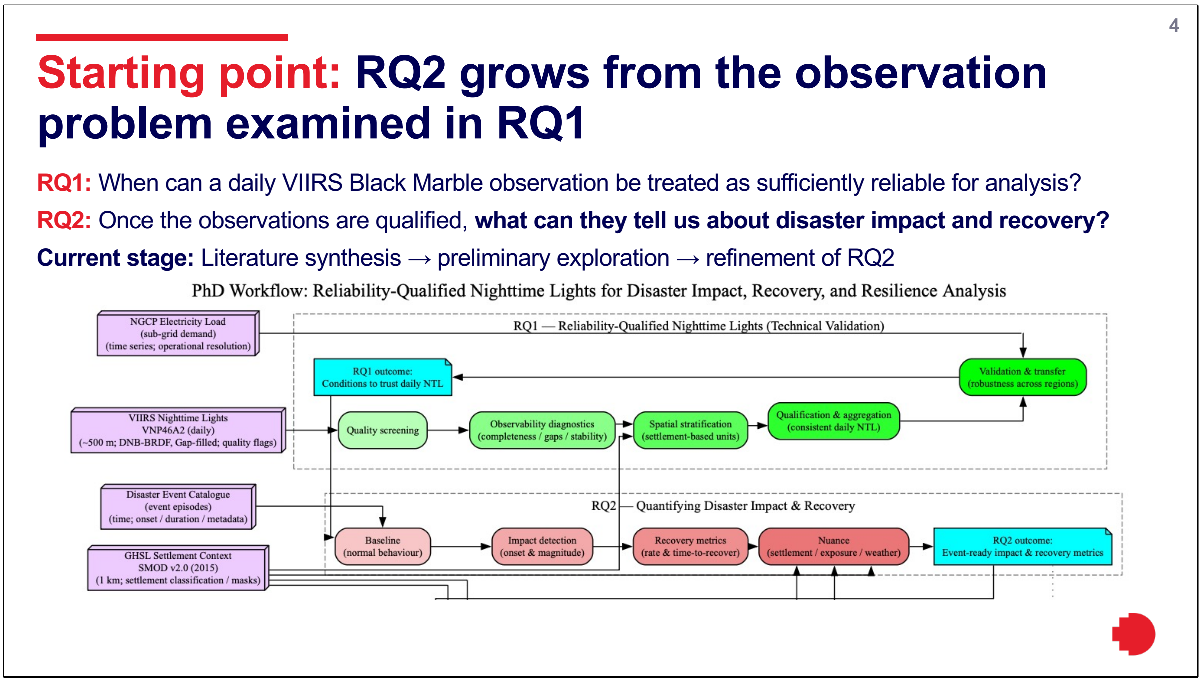
- Existing metrics from literature are crafted under favorable cloud conditions.
- In tropical regions like the Philippines, observability could be as low as 30%.

A recap of what I accomplished so far. RQ1 asked when a daily Black Marble NTL observation can be treated as sufficiently reliable.

The magenta boxes show the input data, while the green boxes show the RQ1 workflow: we did quality screening, observability checks on representative pixel/settlement type, stratified NTL using urban class masks, and validated against electricity load sub-grids.

Using the 2012–onwarss record, we showed that NTL can align well with electricity demand, but only under appropriate settlement masks and valid-pixel thresholds.

The main RQ1 outcome was therefore a set of conditions for trusting daily NTL. The bigger point is that these conditions were established over a long time series; they still need to be tested within the specific phases of a disaster.



RQ2 now zooms into the individual disaster phases shown in the reddish boxes: baseline, impact, recovery, and the possible emergence of a new post-event state.

These phases are not observed under uniform conditions. Before landfall, severe cloud cover may already reduce the observations available for constructing the baseline. Conditions may temporarily clear after the storm, but recovery occurs during the wider typhoon season, when additional storms and persistent cloud can interrupt the record again.

At this finer temporal and spatial scale, viewing geometry may also become more important. A change in VZA between baseline and recovery observations could alter radiance independently of actual functional change.

The emitted signal itself may also evolve. Reconstruction can change buildings, public lighting, fixture types, or the spatial distribution of activity. As Norman noted, local governments may use rebuilding as an opportunity to improve lighting infrastructure, meaning that the post-event signal may not return to the same light-emission regime as before.

So the starting point for RQ2 is to understand how **observation and emission conditions change across each disaster phase**, before interpreting the resulting trajectory as disruption, restoration, or a new normal.

I. What the literature tells us:
Physical and functional recovery are complementary

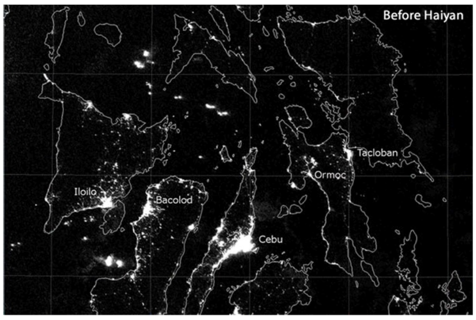
Daytime EO / VHR imagery

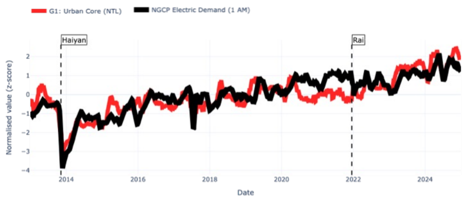
- Strong for debris, damage, reconstruction, land cover, relocation
- Physically specific, but often snapshot-based and cloud/acquisition limited

Nighttime lights

- Strong for electricity-dependent nighttime function and activity
- Higher temporal potential, but semantically less specific

Takeaway:
Physical recovery and functional recovery are related, but not interchangeable.





Ghaffarian’s work shows that different Earth-observation sources capture different recovery proxies: daytime EO and VHR imagery reveal physical damage and reconstruction, meanwhile nighttime lights provide evidence of electricity-dependent nighttime function and activity.

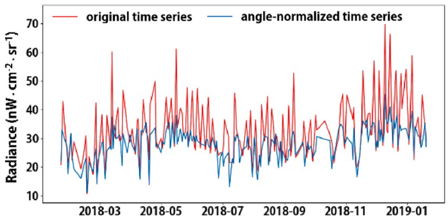
NTL can indicate whether activity continues through restored power, generators, temporary lighting, or changing settlement patterns, even when physical recovery remains incomplete.

The value is therefore in comparing these trajectories—agreement suggests synchronous recovery, while divergence may reveal different recovery processes rather than failure of either proxy.

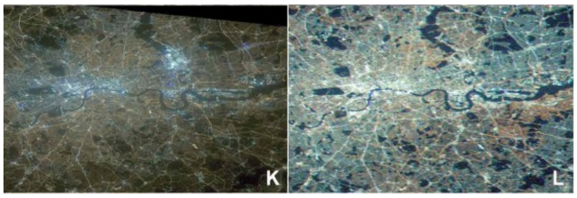
I. What the literature tells us:
RQ1 dealt with clouds; RQ2 zooms into event inference

For RQ2, the event window adds new questions:

- **Baseline choice:** what is “normal” during a pre-Haiyan VIIRS record?
- **VZA / anisotropy:** does geometry alter the apparent impact or recovery?
- **Gap handling:** false precision magnitude arising from interpolation or gap filling
- **Emission regime:** could lighting technology, generators, public lighting, or relocation change the radiance–recovery relationship?



Correcting VZA improves timing and magnitude detection at fine scales (Jia et al., 2023)



LED transition in London captured by DSLR cameras onboard ISS (Sanchez de Miguel et al., 2013)

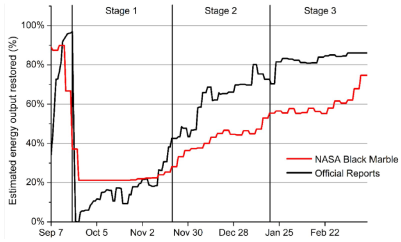
NTL is already established as a disaster-recovery proxy, but once we examine individual event phases, every step—from defining the baseline to detecting impact and interpreting recovery—is shaped by changing clouds, viewing geometry, gap treatment, and the underlying light-emission regime.

This creates a tension between pursuing a technically precise correction framework and keeping RQ2 focused on the applied question of what the disaster trajectory can tell us about functional impact and recovery. The guidance I need from the panel is where to draw that boundary: how much technical control is necessary before the work stops being an application of NTL to recovery and becomes another methodological validation study.

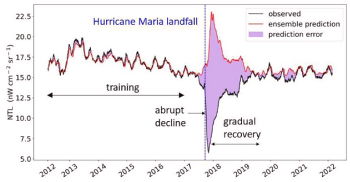
II. From literature gaps to an RQ2 opportunity

Gap synthesis

What is established	What remains unresolved for Haiyan	Proposed RQ2 test
1. Daily Black Marble can track outage and recovery when externally validated *(Román et al., 2019; Mo et al., 2025; Zheng et al., 2025)*	How should the baseline, impact, and recovery phases be defined under phase-varying tropical observability?	**Phase definition and observability audit**
2. Continuity is often created through interpolation, gap filling, compositing, or counterfactual modelling *(Mu et al., 2025; Hao et al., 2023; Tan & Zhu, 2023; Zheng et al., 2026; Thomas et al., 2024)*	How do continuity assumptions change impact depth, impact timing, recovery duration, and longer-term trajectories?	**Observed DNB-BRDF vs gap-filled and modelled trajectories**



Alignment of Black Marble NTL with PREPA load in Puerto Rico after Maria
(Roman et al, 2019)



(b) Hurricane Maria in San Juan

ML-based city-specific NTL projection to quantify disruption
(Chakraborty and Stokes, 2023)

This slide is the transition from the literature review to the proposed RQ2 design. I do not see these as four unrelated gaps; they form a sequence of decisions that must be made before interpreting a disaster trajectory.

First, the literature already shows that daily Black Marble can track outages and recovery when it is validated against electricity or other external evidence. The unresolved question for Haiyan is how to define the relevant disaster phases under tropical conditions. The baseline, immediate impact, early restoration, and longer recovery may each have very different cloud and observation conditions. The first step is therefore to define those phases and establish what was actually available within each one.

Second, many studies address missing observations by filling, interpolating, compositing, or modelling the series. These approaches are useful because machine-learning and counterfactual methods generally require enough continuity to estimate business-as-usual behaviour. They may be particularly valuable for longer-term questions, such as comparing how communities recover over several months. However, filling the series is not neutral. It may smooth the initial decline, move the apparent impact date, shorten the recovery duration, or create a trajectory that appears more certain than the original observations. My proposed test is therefore not to reject modelling, but to quantify what it provides and what it changes by comparing observed DNB-BRDF, the native gap-filled layer, and selected modelled trajectories.

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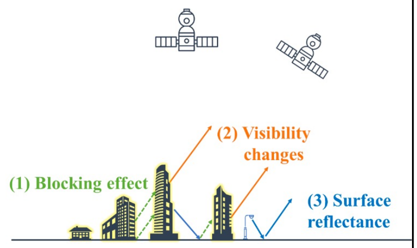
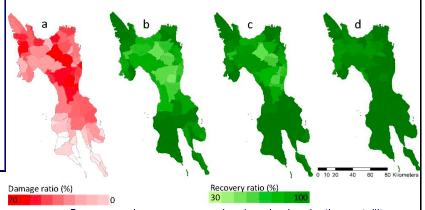
II. From literature gaps to an RQ2 opportunity

Gap synthesis

What is established	What remains unresolved for Haiyan	Proposed RQ2 test
3. Angular response varies with urban morphology *(Li et al., 2019; Tan et al., 2022; Wang et al., 2022; Zheng et al., 2025)*	Is the low-rise and mixed morphology of Samar–Leyte sufficiently angle-sensitive to distort disaster-scale changes?	**Pixel- and settlement-level VZA sensitivity**
4. Earth-observation proxies represent different recovery dimensions *(Ghaffarian et al., 2018; Román et al., 2019; Friedrich et al., 2026)*	When do physical reconstruction and electricity-dependent nighttime function agree or diverge?	**Cross-proxy recovery interpretation**



Varying VZA materiallt changes the NTL radiance (Tan et al, 2021)



Damage and recovery maps developed using daytime satellite imagery via GEE (Ghaffarian et al., 2020)

Prioritize (1) precise reconstruction of the NTL signal, or (2) robust detection & interpretation of functional recovery patterns with targeted sensitivity analyses?

Third, the angular literature shows that VZA effects depend on morphology. Findings from high-rise cities cannot automatically be transferred to Samar–Leyte, where even central business districts are largely low-rise or mixed. I therefore need to test the local relationship rather than assume that VZA is either negligible or universally correctable.

The proposed analysis would examine VZA and radiance at pixel and GHSL settlement levels, then test whether alternative angle restrictions, matching, or correction materially change the expected disaster dip and recovery metrics. The question is not merely whether VZA affects radiance, but whether the effect is large enough to alter the substantive recovery conclusion.

Finally, NTL and physical imagery do not observe the same recovery process. Daytime EO can show debris removal, reconstruction, and land-cover change, while NTL reflects electricity-dependent nighttime function. Where the two trajectories agree, we may observe synchronous physical and functional recovery. Where they diverge, that disagreement may reveal generator use, delayed occupancy, relocation, incomplete service restoration, or another recovery pathway—not necessarily failure of either proxy.

The difficulty is that each of these directions could become a major technical study. My preference is to keep RQ2 application-led: use enough technical control to make the Haiyan recovery interpretation defensible, rather than attempting to develop a universal correction framework for every source of NTL uncertainty.

II. From literature gaps to an RQ2 opportunity

Emerging direction (observability to interpretability)

Working research question

How can reliability-qualified daily VIIRS Black Marble observations be used to characterise disaster impact and recovery **under changing observation and signal conditions**?

Working subquestions

1. How do observability and spatial support vary among the pre-event, impact, and recovery phases?
2. How sensitive are estimated impact and recovery patterns to observation gaps, viewing geometry, spatial support, and baseline choice?
3. How do NTL patterns agree with or diverge from electricity, physical, and contextual recovery evidence?

Main RQ — Why it matters:
Helps ensure that satellite-based recovery claims reflect what affected communities are actually experiencing, rather than artefacts of the observation process.

SQ1 — Why it matters:
Shows decision makers when the satellite has enough evidence to support action—and when cloud or limited coverage means local reports must take priority.

SQ2 — Why it matters:
Prevents technical choices from exaggerating or understating which places were hardest hit or slowest to recover.

SQ3 — Why it matters:
Reveals where restored lighting does—and does not—translate into restored services, rebuilt settlements, and resumed community activity.

III. Proposed first steps
Immediate action plan

The first steps test whether these opportunities are empirically important

- **Event-phase observation audit**
  Describe observation availability during baseline, impact and recovery.
- **VZA sensitivity analysis**
  Test whether comparable viewing conditions change the apparent trajectory.
- **Trajectory and metric sensitivity**
  Compare plausible baselines, spatial supports and recovery summaries.
- **Initial cross-proxy comparison**
  Examine the timing and direction of NTL and available electricity evidence.

These are exploratory steps intended to refine the research question.

The event-phase audit will describe when and where direct DNB-BRDF observations are available. It will examine valid-pixel coverage, observation intervals, quality conditions, and differences among settlement groups.

The VZA analysis will test whether restricting or matching observation angles changes the apparent impact and recovery trajectory. A correction model would only be considered if the exploratory evidence shows that it is necessary and sufficiently stable.

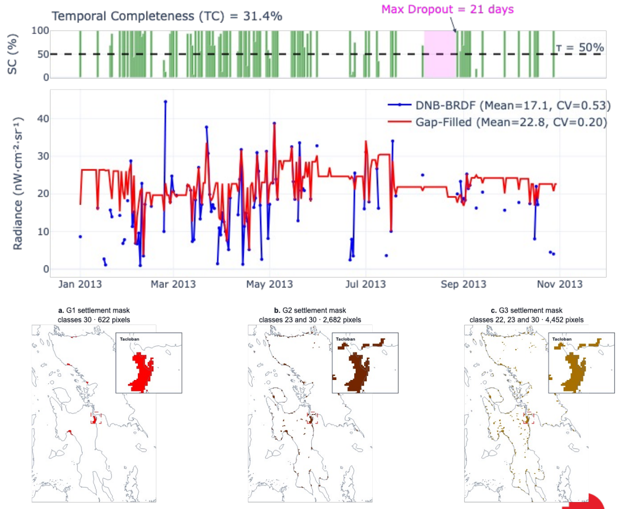
The trajectory analysis will examine whether the main interpretation is sensitive to reasonable baseline, spatial-support, and summary choices. The purpose is initially diagnostic rather than to select a final recovery metric.
The cross-proxy comparison will begin with NGCP data while the suitability of other physical or electricity information is assessed.

IV. Preliminary Haiyan exploration

Exploratory setup

The initial analysis examines how the Haiyan trajectory changes across settlement definitions

- VIIRS VNP46A2 daily Black Marble
- DNB-BRDF as the principal signal
- MQF = 0 for fresh, high-quality observations
- GHSL-SMOD G1–G3 settlement groupings
- NGCP load as an initial external comparison



This first analysis is deliberately simple. I use DNB-BRDF as the main signal, retain only fresh high-quality observations, apply GHSL settlement masks so the signal is not diluted by dark pixels, and use NGCP load as an initial external comparison.

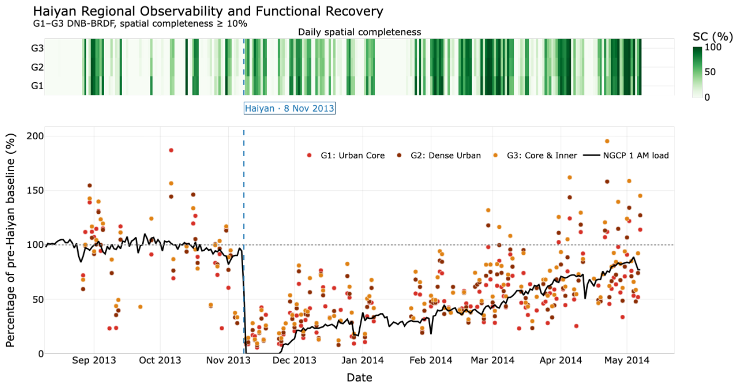
The top plot illustrates why qualification matters: temporal completeness is only about 31%, with some gaps lasting up to 21 days, while the gap-filled layer appears much smoother and more complete. For now, I use a three-month pre-Haiyan period as a provisional reference baseline and a 10% spatial-completeness threshold to retain a daily regional estimate.

Here we to establish what the usable signal looks like under transparent assumptions.

IV. Preliminary Haiyan exploration: Trajectory

NTL trajectory and its observation support should be examined together

- A post-Haiyan radiance decline is visible.
- The immediate direction is broadly consistent with NGCP.
- Observation support becomes uneven across the recovery period.



Here I apply the same setup to the three most urban settlement masks and examine the NTL trajectory together with its observation support.

A clear post-Haiyan decline is visible, and the broad direction is consistent with the NGCP load series. However, the daily NTL values remain highly variable, and the completeness plots show that the baseline, shock, early recovery, and later recovery phases were observed under very different conditions.

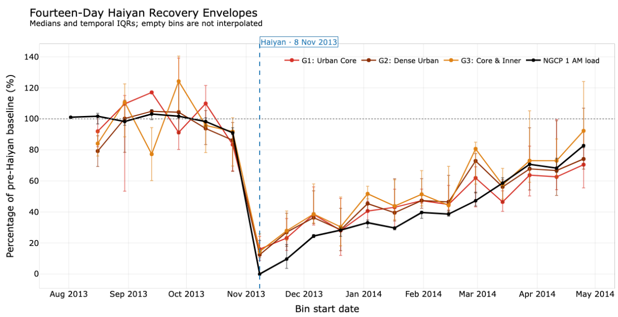
This is already an important result: the event trajectory cannot be interpreted independently of when and how well each phase was observed. What remains unresolved is how much of the fluctuation reflects real functional change and how much reflects the observation process.

IV. Preliminary Haiyan exploration:
Cross-proxy pattern

NTL and NGCP show related but non-identical recovery patterns

- Both series show a major post-impact reduction.
- NTL subsequently rises above the regional NGCP trajectory.
- The source of the divergence is not yet established.

Questions raised: Spatial support? Local restoration? Generators? Activity change? Geometry?



To make the broad trajectory easier to inspect, I aggregate the observations into 14-day bins. This reduces some of the daily volatility and makes the shared post-impact decline and gradual recovery more visible.

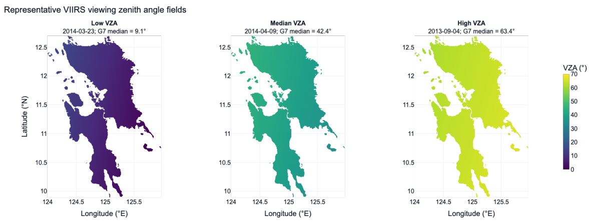
The NTL and NGCP trajectories broadly agree immediately after Haiyan, but later the NTL series rises above the regional load trajectory. That divergence is not yet explained—it could reflect spatial-support differences, local restoration, generators, changing activity, or observation geometry.

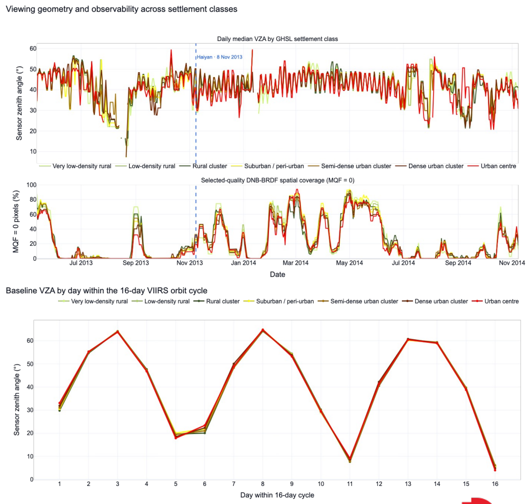
The bigger point is that agreement and divergence are both informative. The task is not to force the proxies to match, but to understand which aspect of recovery each one is capturing.

IV. Preliminary Haiyan exploration:
Geometry pattern

Viewing conditions vary substantially through the event record

- VZA changes through a repeating orbital cycle.
- Near-nadir and off-nadir observations see different spatial configurations.
- Clouds and quality screening leave an uneven subset of the cycle.





This slide examines the viewing conditions behind the trajectory.

VZA changes through the repeating VIIRS orbital cycle, so the same location is not observed from the same angle every night. Clouds and quality screening then remove parts of that cycle, leaving an uneven subset of near-nadir and off-nadir observations across the event period.

The three maps illustrate how different the same region looks under low, medium, and high VZA. At this stage, the key question is whether these geometry changes are large enough to alter the apparent disaster signal, rather than simply whether VZA varies.

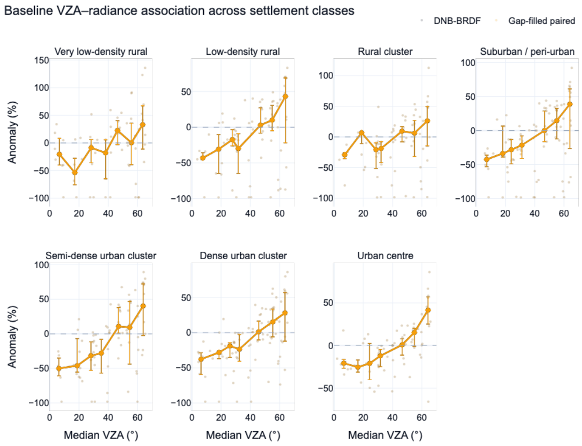
IV. Preliminary Haiyan exploration:
VZA association

Within-pixel analysis suggests that VZA may warrant further testing

Exploratory result: Selected specifications suggest approximately 4–9% radiance change per additional 10° of VZA.

Still unresolved

- Stability across time and settlement types
- Role of azimuth and other conditions
- Whether the effect changes the event interpretation
- Appropriate treatment: matching, restriction or correction



The within-pixel baseline analysis suggests a possible VZA–radiance association across several settlement classes. Under the selected specifications, radiance changes by roughly 4–9% for each additional 10 degrees of VZA.

This is large enough to warrant further testing, but it is not yet a correction factor. The relationship still needs to be tested for temporal stability, settlement dependence, azimuth effects, and whether it materially changes the estimated impact or recovery pattern.

The practical decision is therefore whether VZA should be handled through matching, restriction, correction, or simply carried forward as an uncertainty flag. The priority is not to correct geometry for its own sake, but to determine whether it changes the substantive recovery interpretation.

V. Buring Questions (for discussion)

Question	Current thinking	Why I need guidance
1. What should be the central RQ2 contribution?	An application-led study of Haiyan impact and functional recovery, supported by reliability and sensitivity tests.	The technical issues could easily become the main study and repeat the validation emphasis of RQ1.
2. How should disaster phases and physical-recovery evidence be integrated?	Define baseline, shock, early recovery, later recovery, and possible new normal; compare NTL with available reconstruction and settlement evidence.	Phase boundaries are not universal, and physical and functional recovery may occur at different times and scales.
3. How far should the VZA analysis go?	Quantify local VZA sensitivity by pixel and settlement type, then test whether it changes the inferred impact or recovery pattern.	A full angular-correction framework may become a separate methodological paper without improving the substantive interpretation.

I want to end by making the decisions I am currently facing explicit, rather than presenting the workflow as already fixed.

First, I need advice on the central contribution. My preference is for RQ2 to remain application-led: understanding Haiyan's functional impact and recovery, while using observability, VZA, and gap sensitivity as supporting controls. My concern is that these technical components could expand until RQ2 becomes another reliability-validation study.

Second, I need guidance on defining the disaster phases and connecting them with existing Haiyan physical-recovery evidence. The literature uses different phase definitions, and the appropriate boundaries may depend on both the event process and the observations available. Physical reconstruction and functional activity may also recover asynchronously, so the comparison should not assume that the two trajectories must match.

Third, the initial VZA results suggest that geometry may affect radiance enough to warrant testing. My current view is to quantify the local effect and determine whether it changes the impact or recovery interpretation. I am unsure whether that is sufficient, or whether the study requires a more formal angular correction.

V. Buring Questions (for discussion)

Question	Current thinking	Why I need guidance
4. What level of external validation is sufficient?	Use NGCP load as the main functional comparison, supplemented where possible by outage, restoration, facility, or local evidence.	NGCP represents regional transmission demand, not household-level electricity access or local distribution recovery.
5. Which recovery outcomes are most meaningful?	Prioritise impact magnitude, duration of disruption, recovery trajectory, and spatial differences among settlements.	I need to decide whether the output should be an established outage metric, a recovery classification, or identification of communities requiring further investigation.
6. Who is the primary audience?	An application-led paper for disaster-recovery and Earth-observation researchers, with practical relevance to utilities and disaster agencies.	The journal and framing depend on whether the main contribution is technical reliability or applied recovery interpretation.

Fourth, NGCP load provides a valuable regional functional comparison, but it does not directly measure household reconnection or local distribution recovery. I need advice on whether this level of validation is adequate, or whether additional local restoration evidence is essential for the claims I want to make.

Fifth, I need to decide what the most useful recovery output should be. It could be a metric such as disruption duration or time to recovery, but it could also be a spatial identification of settlements where recovery appears delayed, uncertain, or inconsistent with other evidence. The latter may be more meaningful for decision makers, but it requires careful interpretation.

Finally, the target audience depends on where this balance lands. I am more interested in an applied disaster-recovery contribution—particularly what NTL can reveal for storm-affected communities that are poorly represented by methods developed under more favourable observation conditions. However, if the technical uncertainty analysis becomes dominant, the paper may need to be framed primarily for the remote-sensing methods community.

The overarching question: **how much technical refinement is necessary to make the recovery interpretation defensible without allowing the technical problem to replace the disaster-recovery question?**

Progress to date and immediate priorities

RQ1 completed and progressing to publication

- Presented and published in the **ISPRS Archives,** full first-paper draft submitted for review

New funding and transferability opportunity

- Awarded **NHRA scholarship funding.** Future consideration: transferability in an Australian hazard context Retain the broader argument that the framework is relevant to cloud-impacted and data-scarce regions

RQ2 immediate priorities

- Continue targeted literature review and refine the concept note
- Extend Haiyan exploratory analysis across phases, gaps, VZA, and cross-proxy evidence
- Resolve the central scope and contribution
- Identify the intended users, target journal, and publication framing early

Cotutelle administration

- Confirm supervision and meeting arrangements
- Clarify programme milestones, documentation, and upcoming requirements

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